MIL-C-25101A(USAF)

5 December 1956

Superseding
MIL-C-25101(USAF)
13 January 1955

MILITARY SPECIFICATION

CONTROL, SLAVING; N-1 COMPASS

1. SCOPE

1.1 This specification covers one type of an airborne slaving control, for use with the Type N-1 Compass.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification.

SPECIFICATIONS

Military

MIL-N-6011	Nitrogen; Liquid and Gas
MIL-C-8780	Compass, Slaved, Directional Gyro Stabilized, Type N-1

STANDARD

Air Force-Navy Aeronautical

AN3102	Connectors-Electrical, Receptacles, Box Mounting

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction.- This specification makes provisions for preproduction testing.

3.2 General Specification.- The requirements of MIL-C-8780 apply as requirements of this specification with additions specified herein.

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3.3 Design and Construction.- The slaving control shall be designed to operate as a rate gyro sensitive to rotation about its vertical axis. The configuration of the slaving control shall conform to figure 1.

3.3.1 Hermetic Seal.- The slaving control shall be hermetically sealed in nitrogen at an absolute pressure of 1 ±1/15 atmosphere. The nitrogen shall conform to Specification MIL-N-6011, Type I, grade A. Provisions for compensation and necessary control shall be provided through the seal.

3.3.2 The case shall be legibly and permanently marked with the direction of flight and shall be provided with a flange suitable for mounting in an aircraft as shown in figure 1.

3.3.3 Power Requirements.- The slaving control shall operate on 27.5V dc power and 115 volts, 3-phase, 400 cps, ac power and shall not exceed 2.0 watts dc and 24 volt-amperes alternating current.

3.3.4 Power Connectors

3.3.4.1 Type N-1 Compass Interconnection.- The slaving control shall be equipped with a Cannon GS02-20-15P connector or its equivalent, to provide for interconnection with the AN3102A-20-15S connector on the amplifier of the Type N-1 compass. In the GS02-20-15P receptacle, the control circuit shall be connected to pins "E" and "F". Power shall be applied to the slaving control through the following pins in the GS02-20-15P connector:

PINS	
A	A phase, 115V, 400 cps, 27.5V dc GND
B	B phase, 115V, 400 cps
C	C phase, 115V, 400 cps
D	+27.5V dc

3.3.4.2 Other Equipment Interconnection.- The slaving control shall be equipped with a Cannon GS02-18-1P receptacle, or its equivalent for interconnection with other equipment. In the GS02-18-1P receptacle, the control circuits shall be connected such that pin "A" is common for pins "B" and "C" and pin "D" is common for pins "E" and "F".

3.4 Performance.- The slaving control shall meet the performance requirements specified herein and shall perform satisfactorily after being subjected to the tests specified in section 4.

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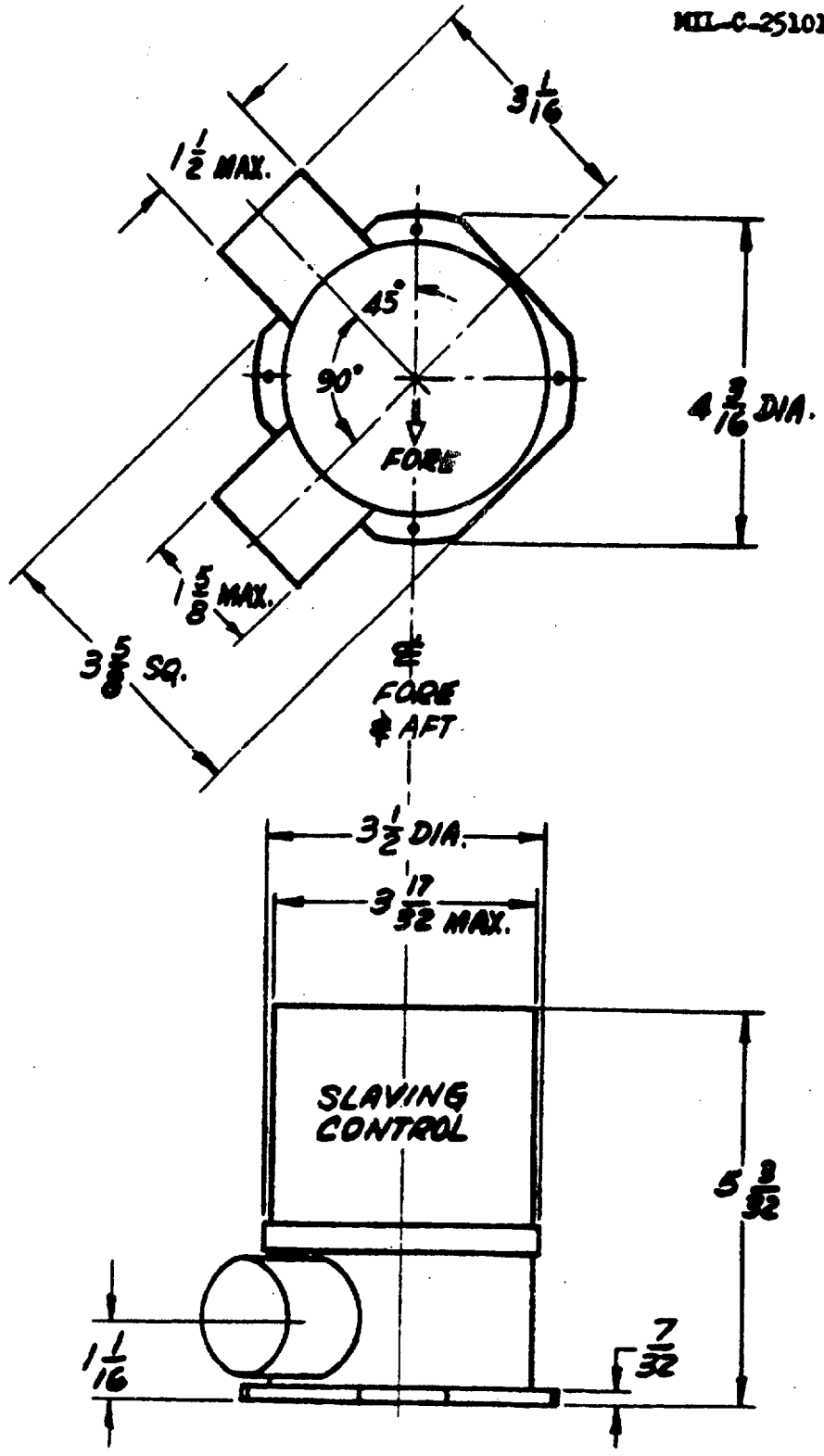


FIGURE 1

SLAVING CONTROL OUTLINE

TOLERANCES:
FRACTIONS ± 1/32
ANGLES ± 1°

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3.5 Equipment Characteristics

3.5.1 Control Relays.- The slaving control shall contain left and right turn contacts electrically isolated from each other which shall respectively control 2 poles on the left and right turn double-throw relays, and whose points shall individually terminate at the Cannon GS02-18-1P receptacle, or its equivalent.

3.5.1.1 Sensitivity.- The slaving control shall operate to close the slaving relays in a minimum rate of turn of 25 ±2 degrees per minute in either right or left turning directions. The slaving relays shall remain closed during the remainder of the turn.

3.5.2 Rate Gyro

3.5.2.1 Operating Speed.- The time for the gyro motor to reach operating speed shall not exceed 10 minutes.

3.5.2.2 Freedom of Gyro Movement.- The freedom of the gyro movement between contacts shall be a minimum of plus or minus 3.5 degrees.

3.5.2.3 Gyro Stops.- The gyro shall be limited by stops at a rate of 60 ±5 degrees per minute in either direction of turn.

3.5.2.4 Direction of Precession.- The direction of precession shall be such that in a right turn the right turn contact is closed and is the contact which is closed by a momentary change in pitch in the climb direction.

3.5.2.5 Freedom of the Vertical Gimbal.- The freedom of the vertical gimbal shall be limited by flexible stops to not more than plus or minus 6 degrees and solid stops to not less than plus or minus 8 degrees.

3.5.2.6 Damping.- The damping of the vertical gimbal shall be of a degree which will permit compliance with the requirements herein.

3.6 Weight.- The weight of the slaving control shall not exceed 4-1/4 pounds.

3.7 Government Loaned Property.- When provided for in the contract, the following items will be furnished by the Government on loan to the contractor, upon his request:

USAF Stock No.	Description	Quantity
6262 - 15810	Directional Control	1
6262 - 16000	Amplifier	1
6263 - 15900	Master Indicator	1
6240 - NSL	DT-173/AJN Detector	1
6263 - 15150-1	Slaving Control	1

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QUALITY ASSURANCE PROVISIONS

4.1 General.- The inspection and testing of the Control, Slaving, N-1 Compass shall conform to section 4 of Specification MIL-C-8780. Additional requirements are specified herein.

4.2 Acceptance Tests.- Acceptance tests shall consist of:

a. Individual tests
b. Sampling plans and tests

4.2.1 Individual Tests.- Each slaving control shall be subjected to the tests as specified herein.

4.2.1.1 Freedom of Movement.- The slaving control shall be properly connected to test apparatus and set into operation on a Scorsby machine (or equivalent roll, pitch and yaw testing apparatus which shall provide 7 to 8 oscillations per minute) which oscillates in yaw, pitch, and roll through an angle of 2.5 degrees on either side of the reference axis with a period of 8 seconds per oscillation. The slaving control relays shall not close after the test has been in progress for one minute. The test shall be run for a minimum period of time of 3 minutes. A momentary make of contact during this test shall not be cause for rejection.

4.2.1.2 Rate of Turn and Oscillation Tests.- The slaving control shall be mounted and operated on a combination rate of turn and oscillation table and subjected to a continuous oscillation of plus or minus 1 degree amplitude with a period of 2 seconds and a rate of turn of 45 degrees per minute. One minute after start of tests the contacts of the slaving control shall be closed and shall not open at any time during the tests. The tests shall be run for a minimum period of time of 3 minutes.

4.2.1.3 Operating Speed.- The time for the slaving control to reach operating speed and to erect to its normal position shall be noted and shall not exceed 3 minutes. The gyro has reached operating or synchronous speed when the ammeter reading connected on the "B" phase suddenly drops in value as opposed to a gradual decrease. The current reading after 10 minutes shall not exceed 0.15 amperes.

4.2.1.4 Oscillation Table Tests.- The slaving control shall be mounted and set into normal operation on an oscillation table having an amplitude of plus or minus 4 degrees and a period of 2 seconds. One minute after start of tests the contacts shall be open and shall not close during tests. The tests shall be run for a minimum period of time of 3 minutes.

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4.2.1.5 Rate Table Tests.- The slaving control shall be properly connected and set into operation on a rate table. After turning 45 degrees at a rate of 45 degrees per minute, the turn shall be stopped and the time for the relays to open shall be 4 ±1 second. After turning 45 degrees at a rate of 90 degrees per minute, the turn shall be stopped and the time for the relays to open shall be 6 ±2 seconds.

4.2.1.6 Dielectric.- Dielectric tests shall be made between isolated pins and between pins and case, where no circuit is closed, for a period of 10 seconds. There shall be no breakdown of insulation. The following dc potentials shall be applied: 280V dc between motor leads and case, 500V dc between all other pins and case.

4.2.2 Sampling Plans and Tests

4.2.2.1 Sampling Plan.- Five slaving controls shall be selected for testing in accordance with Specification MIL-C-8780.

4.2.2.2 Sampling Tests.- Sampling tests on slaving controls shall be conducted by combining the slaving control with the other components of the Type N-1 compass and submitting the complete compass to the sampling tests specified in Specification MIL-C-8780.

4.3 Preproduction Testing

4.3.1 Preproduction Test Samples.- Three slaving controls shall be subjected to the preproduction tests of Specification MIL-C-8780.

4.3.2 Preproduction Tests.- Preproduction tests on the slaving control shall be conducted by combining the slaving control with the other components of the Type N-1 compass and submitting the complete compass to the preproduction tests specified in Specification MIL-C-8780.

5. PREPARATION FOR DELIVERY

5.1 General.- All preservation, packaging, packing, marking, and testing shall conform to the requirements of Specification MIL-C-8780.

6. NOTES

6.1 Intended Use.- The Control, Slaving; N-1 Compass covered by this specification is intended for use in all aircraft whose range of operation includes the polar regions as well as all aircraft requiring precise heading reference stabilization.

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6.2 Definitions

6.2.1 Amplitude.- The word "amplitude", as specified herein shall mean the extent of motion as measured from one extreme to the opposite extreme.

6.3 Ordering Data.- Procurement documents should specify the following.

6.3.1 Preproduction Tests.- It is expected that the contract or purchase order will specify that three slaving controls will be required as preproduction samples and that these preproduction samples will be subjected to the preproduction tests to determine compliance with the requirements of this specification. The invitation for bids and the contract should specify the point of inspection for these tests.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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